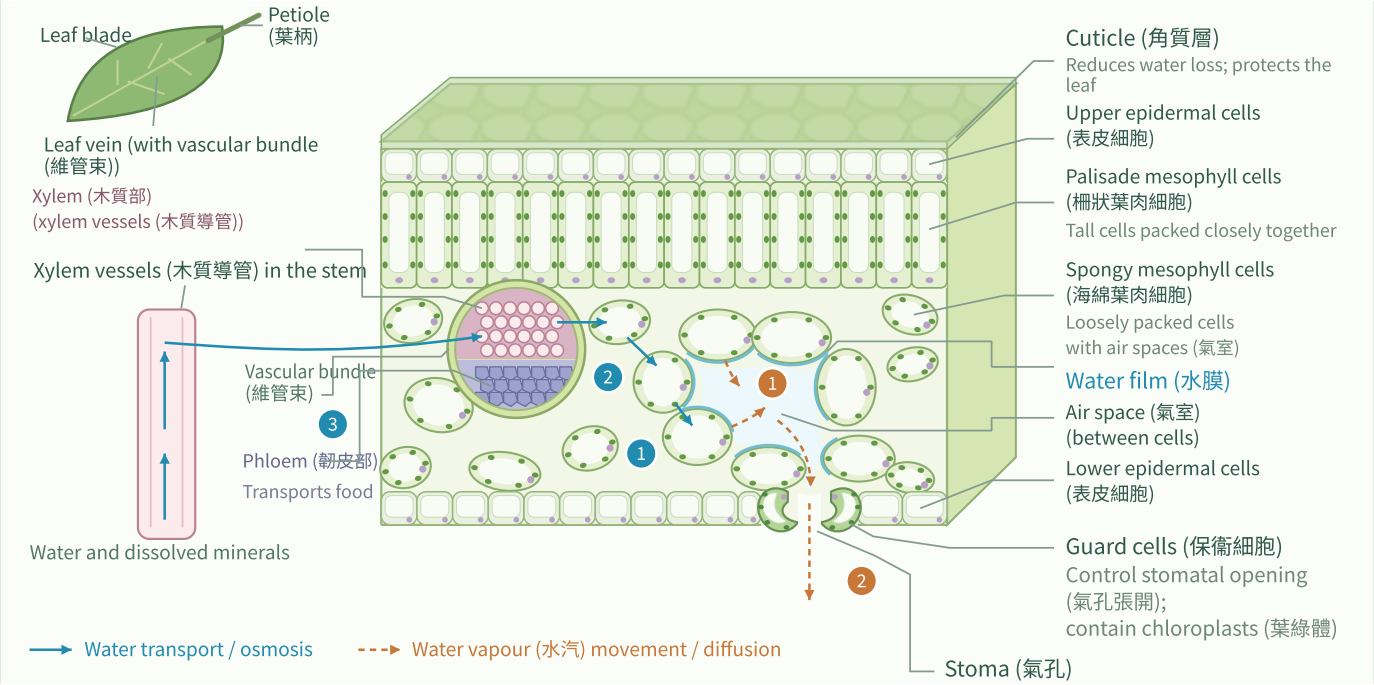
Name: _____ Class and student number: _____

Give out C only after the original investigation answers are fixed.

Learn about transpiration (蒸騰) from the red water trace



Transpiration (蒸騰) through stomata (氣孔)

1 Water evaporates (蒸發) from the surfaces of mesophyll cells (葉肉細胞) into the air spaces (氣室).

2 Water vapour (水汽) in the air spaces (氣室) diffuses through stomata (氣孔) into the air.

How transpiration pull (蒸騰拉力) forms

1 Mesophyll cells (葉肉細胞) lose water, so their water potential falls.

2 Mesophyll cells (葉肉細胞) take in water from nearby cells by osmosis.

3 Water is drawn out of the xylem vessels (木質導管), creating transpiration pull (蒸騰拉力).

1. Transpiration (蒸騰) is the process in which water in a plant evaporates (蒸發) and is lost to the air as water vapour (水汽). Water vapour (水汽) mainly leaves through stomata (氣孔) in leaves and herbaceous stems (草本莖).
   a. Water evaporates (蒸發) from the water film (水膜) on the surface of mesophyll cells (葉肉細胞) and enters the air spaces (氣室) in the spongy mesophyll as water vapour (水汽). When the water vapour (水汽) concentration in the air spaces (氣室) is higher than in the air outside, water vapour (水汽) leaves the leaf through stomata (氣孔) by diffusion.
   b. When mesophyll cells (葉肉細胞) lose water, their water potential falls. They take in water from nearby cells by osmosis. This draws water out of the leaf's xylem vessels (木質導管) and creates transpiration pull (蒸騰拉力), which pulls water upwards through the stem's xylem vessels (木質導管).
2. Under suitable conditions, such as enough water being available, higher light intensity (光強度) usually makes stomata (氣孔) open wider. Water vapour (水汽) then leaves the leaf faster by diffusion, so the transpiration rate (蒸騰速率) increases.
   Note: More light may also make the leaf's temperature rise, increasing transpiration (蒸騰) further. This investigation therefore assumes that all setups have the same temperature so we can compare the effect of light intensity (光強度).
3. Transpiration pull (蒸騰拉力) keeps roots taking in water and moves water and dissolved minerals absorbed by the roots upwards through xylem vessels (木質導管) to the leaves.

See textbook section 11.1, pages 11-3 to 11-8. The red water trace gives evidence of water transport. Its upward speed is not a direct measure of the transpiration rate (蒸騰速率). Comparing celery with and without leaves alone also cannot prove the role of stomata (氣孔).

Internal leaf structure and transpiration (蒸騰)

Orange numbers: transpiration (蒸騰) through stomata (氣孔) | Blue numbers: how transpiration pull (蒸騰拉力) forms

Enlarged view, not to scale. The celery stalk in this experiment is a petiole (葉柄). The stem shown here explains transport through xylem vessels (木質導管).

Return to your first ideas

Refer to the original hypothesis and reason in A. You do not need to rewrite them from memory.

Do your data support your original hypothesis? Use ideas from the key learning points above to improve your original explanation. Explain how it relates to your data.